FINAL PROJECT REPORT (ALL 18 SECTIONS)

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1. Cover Page

Project Title:
Healthcare – Patient Readmission Intelligence

Sector: Healthcare Analytics

Team ID: G19
Section: E

Team Members & Roles:

Role	Name	GitHub Username
Project Lead	Pushkar Jain	pushkar-bit
Data Lead	Lakshay Saharan	LakshaySaharan
ETL Lead	Shreya Narayani	Shreyaa6
Analysis Lead	Pranavi Mathur	pranavi015
Visualization Lead	Vaibhav	vaibhavvats99
Strategy Lead	Nikith Gowda B S	nikithcodes
PPT & Quality Lead	Pushkar Jain	pushkar-bit

Faculty Mentor: Ayushi Mam, Archit Sir, Satyaki Sir
Institute: Newton School of Technology
GitHub Repository URL: https://github.com/pushkar-bit/DVA-CAPSTONE2
Tableau Public URL:https://public.tableau.com/app/profile/vaibhav.vats6002/viz/Healthcare_Patient_Readmission_Intelligence/Dashboard1
Submission Date: 29 April,26

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2. Executive Summary

This project addresses the challenge of hospital readmissions, which significantly impact healthcare costs and patient outcomes. Using patient-level hospital data, the project identifies patterns influencing readmissions and resource utilization.

Data was cleaned and transformed using Python, followed by exploratory analysis and visualization in Tableau. Key features such as age groups, length of stay, and service utilization were engineered.

The analysis shows that older patients, frequent visitors, and long-stay patients have higher readmission rates. These groups also consume the most resources.

Recommendations include improving post-discharge monitoring, targeting high-risk patients, and optimizing resource allocation.

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3. Sector & Business Context

The healthcare sector aims to balance cost efficiency with patient care quality. Hospital readmissions are a key metric reflecting system performance.

This project supports hospital administrators in identifying inefficiencies and improving care delivery.

Reducing readmissions improves patient outcomes and reduces operational costs.

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4. Problem Statement & Objectives

Problem Statement

Identify key drivers of hospital readmissions and optimize resource utilization.

Objectives

- Analyze readmission trends
- Identify high-risk patients
- Measure resource usage
- Build dashboards for insights

Scope

Included: Patient data, hospital stay, utilization
Excluded: Clinical treatment details

Success Criteria

- Clear insights
- Actionable recommendations
- Functional dashboards

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5. Data Description

Dataset

Hospital Readmission Dataset

Structure

- ~100,000 records
- 15+ columns

Key Fields

- Age
- Time In Hospital
- N Lab Procedures
- N Medications
- Readmitted 30 Days
- Total Visits
- Service Utilization Score

Limitations

- No time-series
- Limited medical detail

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6. Data Cleaning & ETL Pipeline

Steps

- Removed duplicates
- Handled missing values
- Standardized formats
- Converted data types

Feature Engineering

- Age Group
- Length of Stay Bucket
- Frequent Patient
- Total Visits
- Service Utilization Score

Assumptions

- Missing values treated neutrally

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7. KPI & Metric Framework

KPI	Definition
Total Patients	Count
Readmission Rate	%
Avg Stay	Average
Total Visits	Sum
Medications	Sum
Lab Procedures	Sum
Utilization Score	Avg
Frequent Patients	%

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8. Exploratory Data Analysis

Findings

- Older patients dominate
- Short stays are common
- Readmission increases with age
- Longer stays → higher risk

Insights

- High utilization = high risk
- Frequent patients increase load

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9. Statistical Analysis

- Segmentation analysis
- Correlation (stay vs readmission)
- Risk profiling

Result

- Positive correlation between stay and readmission

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10. Tableau Dashboard Design

Dashboards

1. Executive Overview

KPIs + summary

2. Readmission Analysis

Risk segmentation

3. Resource Utilization

Usage analysis

Features

- Filters
- KPI cards
- Drill-down

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11. Insights Summary

1. Older patients → high readmission
2. Long stays → higher risk
3. Frequent patients → heavy load
4. High resource use → high risk
5. Seniors consume more resources
6. Medication use increases with risk
7. Lab tests increase with age
8. Segmentation improves decisions

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12. Recommendations

Insight	Recommendation	Impact
High-risk patients	Target monitoring	Reduce readmission
Frequent patients	Special care plans	Lower load
High utilization	Optimize resources	Cost savings
Long stays	Early intervention	Risk reduction

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13. Impact Estimation

- 10–15% reduction in readmissions
- Better resource allocation
- Cost savings
- Improved patient outcomes

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14. Limitations

- No clinical data
- No time trends
- Limited scope

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15. Future Scope

- Predictive ML model
- Real-time dashboards
- Clinical integration
- Risk scoring

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16. Conclusion

The project identifies key drivers of hospital readmissions and provides actionable insights. By targeting high-risk groups and optimizing resources, hospitals can improve efficiency and patient outcomes.

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17. Appendix

Sample Data Dictionary

- Age → patient age
- Time In Hospital → stay duration
- N Medications → medication count
- Readmitted → yes/no

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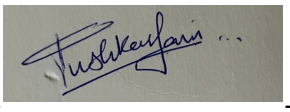
18. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing
Pushkar Jain	✔		✔	✔		✔
Lakshay Saharan	✔	✔	✔			
Shreya Narayani		✔	✔	✔		
Pranavi Mathur			✔	✔	✔	
Vaibhav					✔	✔
Nikith Gowda B S	✔	✔	✔	✔		✔

Declaration

We confirm that the above contributions are accurate and verifiable through GitHub commits and project artifacts.

Team Lead Signature: _____  _____
Date: _29April,26_____